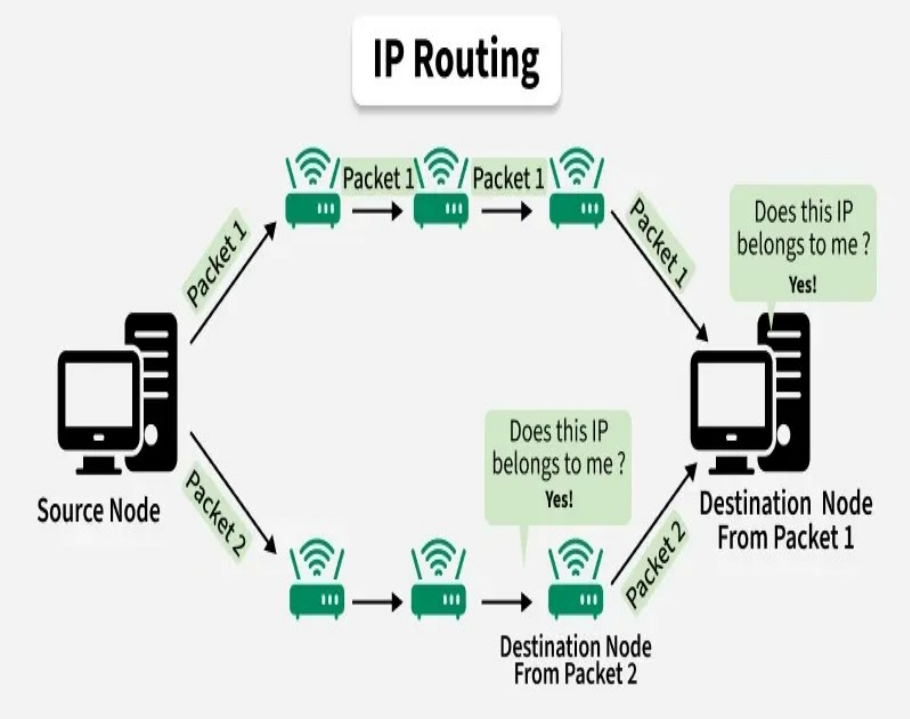
Routing Overview

Routing Overview

- Network routing is the process of selecting the best path for data to travel across one or more networks.
- It plays a vital role in the internet by ensuring that data packets reach their correct destination efficiently.
- Routing determines the path IP packets follow from source to destination.
- It is used in packet-switched networks like the Internet.
- Different routing methods are used to optimize speed, reliability, and performance.

Routing of a data packet is done by analyzing the destination IP Address of the packet



- The Source Node (Sender) sends the data packet on the network, embedding the IP in the header of the data packet.
- The nearest router receives the data packet, and based on some metrics, further routes the data packet to other routers.
- Step 2 occurs recursively till the data packet reaches its intended destination.
- **Note:** There are limits to how many hop counts a packet can do if it is exceeded, the packet is considered to be lost.

Different Types of Routing

Routing is typically of 3 types, each serving its purpose and offering different functionalities:

- Static Routing
- Dynamic Routing
- Default Routing

Static Routing

Static routing is a non-adaptive routing method where routes are configured manually by the network administrator:

- It provides complete control over routing decisions but is best suited for small networks.
- Routes are added manually
- Full control over packet routing
- Simple but hard to manage in large networks

Static Routing

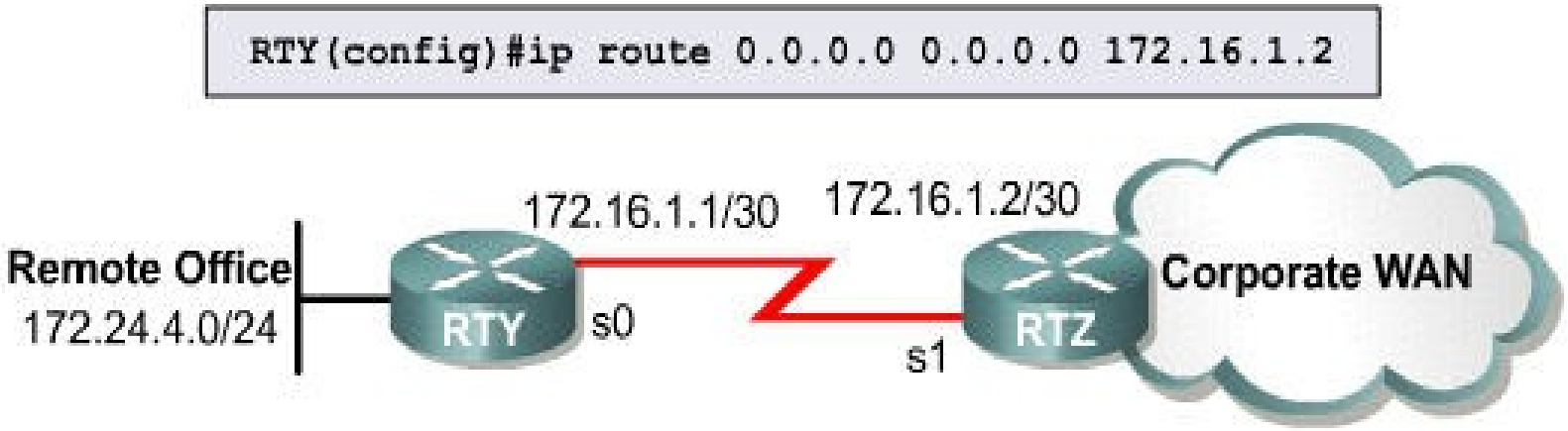
- Static routing is **useful in networks that have a single path to any destination network.**

- Static routing **reduces the memory and processing burdens on a router**.

- To configure static routing on a Cisco router, use the **`ip route`** command. This command uses the following syntax:

```
Router(config)#ip route destination-network
destination-netwok-mask {next-hop-ip
address | interface} [distance] [tag tag]
[permanent]
```

Configuring Static Default Routes



Advantages and Dis-Advantages of Static Routing

Static Routing Advantages	Static Routing Disadvantages
• Low processor overhead. Routers do not spend valuable CPU cycles calculating the best path. This requires less processing power and less memory and therefore, a less expensive router. • No bandwidth utilization. Routers do not take up bandwidth updating each other about static routes. • Secure operation. Routers that do not send updates will not inadvertently advertise network information to an untrusted source. Routers that do not accept routing updates are less vulnerable to attack. • Predictability. Static routes enable an administrator to precisely control the router path selection. Dynamic routing sometimes yields unexpected results, even in small networks.	• High maintenance configuration. Administrators must configure all static routes manually. Complex networks may require constant reconfiguration. • No adaptability. Statically configured routes cannot adapt to changes in link status

Dynamic Routing

Dynamic routing is an automatic and adaptive routing method where routers choose paths using algorithms.

It is widely used in modern networks due to its flexibility.

- Routes update automatically
- Adapts to network changes
- Suitable for large networks

Dynamic Routing

- Dynamic routing of TCP/IP can be **implemented using one or more protocols which are often grouped according to where they are used.**

- **Routing protocols designed to work inside an autonomous system** are categorized as **interior gateway protocols (IGPs).**

- **Protocols that work between autonomous systems are classified** as **exterior gateway protocols (EGPs).**

- Protocols can be **further categorized** as **either distance vector or link-state routing protocols**, **depending on their method of operation.**

Advantages and Dis-Advantages of Dynamic Routing

Dynamic Routing Advantages	Dynamic Routing Disadvantages
• High degree of adaptability. Routers can alert each other about links that are down or about newly discovered path. Routers automatically learn the topology of a network and select optimum paths.	• Increased processor overhead and memory utilization. Dynamic routing processes can require a significant amount of CPU time and system memory.
• Low maintenance configuration. After the basic parameters for a routing protocol are set correctly, administrative intervention is not required.	• High bandwidth utilization. Routers use bandwidth to send and recieve routing updates, which can detrimentally affect performance on slow WAN links.
• Routers can alert each other about links that are down or about newly discovered path(s).	

Default Routing

Default routing sends packets to a predefined gateway when no specific route is available. It is commonly used in networks with a single exit point.

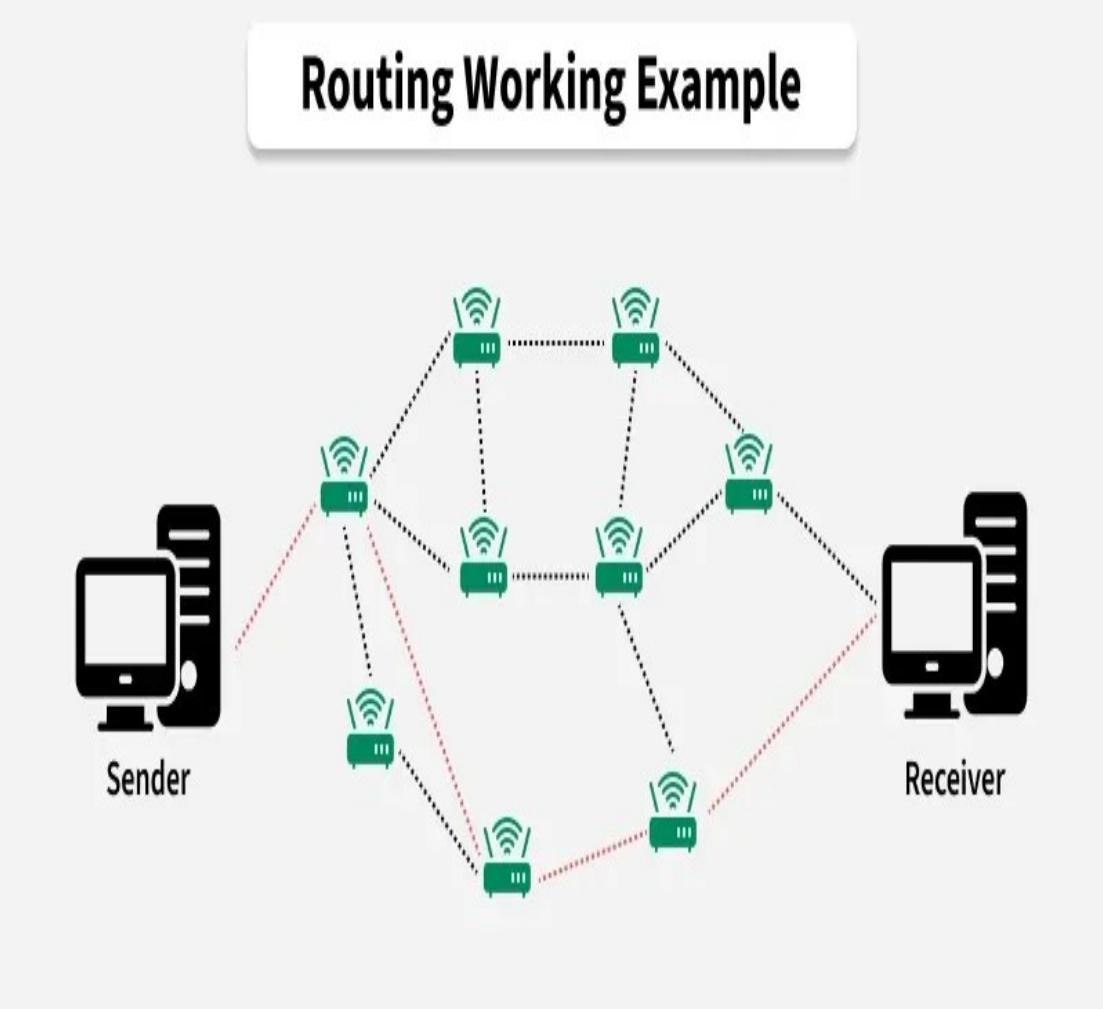
- Used when no route is found
- Simple and efficient setup
- Default route: 0.0.0.0/0

Working Principle of Routing

Routing works by finding the shortest path from the source node to the destination node across a network. Here's the step-by-step working of routing:

- **Step 1: Communication Start -** A device (client or server) starts communication using the HTTP protocol.
- **Step 2: Data Packets**: The data is broken into small packets. Each packet is tagged with the destination IP address.
- **Step 3: Routing Table -** Routers use a routing table to find the best and shortest path to the destination.
- **Step 4: Hopping -** Packets travel through multiple routers (hops) to reach the destination. There is a limit on the number of hops; if exceeded, the packet is dropped and retransmitted.
- **Step 5: Destination -** Packets reach the destination, are reassembled, and checked for errors.

Routing Working Examples



In this image, we have 3 major components:

- Sender
- Receiver
- Routers

The shortest path is highlighted in red, the path with the least hop count.

As we can see, there are multiple paths from source to node but if all the appropriate metrics are satisfied, the data packets will be transmitted through the shortest path (highlighted in red).

Main Routing Protocols

- **RIP (Routing Information Protocol):** It is a distance-vector protocol that uses hop count as a metric.

- **OSPF (Open Shortest Path First):** OSPF is a link-state protocol that finds the shortest path using the Dijkstra algorithm.

- **EIGRP (Enhanced Interior Gateway Routing Protocol):** It is a hybrid protocol that combines features of distance-vector and link-state.

- **BGP (Border Gateway Protocol):** It is a path-vector protocol that is used for routing between different autonomous systems on the internet.

- **IS-IS (Intermediate System to Intermediate System):** It is a link-state protocol that is primarily used in large networks like ISPs.

Methods of operation for Routing Protocols

- The purpose of routing protocols is to learn about all the available paths to route data packets, build routing tables, and make routing decisions based on specified metrics.

- There are two primary Operations of routing protocols

Distance Vector Routing

- In this type of routing protocol, all the nodes that are a part of the network advertise their routing table to their adjacent nodes (nodes that are directly connected) at regular intervals.
- With each router getting updated at regular intervals, it may take time for all the nodes to have the same accurate network view.
- Uses fixed length sub-net, not suitable for scaling.
- Algorithm used: Bellman Ford Algorithm to find the shortest path.

Link State Routing

- Link State Routing is a type of dynamic routing protocol where routers send updates only when there is a change in the network, saving bandwidth.
- Routers share information about links such as cost and hop count to find the best path.
- It uses Variable Length Subnet Mask (VLSM) for better and scalable addressing.
- It uses Dijkstra's Algorithm to calculate the shortest path.

Summary

Routing Protocols	Method of Operation
AppleTalk RTMP	Distance vector protocol
EIGRP	Advanced distance vector protocol (sometimes referred to as a "hybrid" protocol)
IGRP	Distance vector protocol
IPX RIP	Distance vector protocol
IS-IS	Link-state protocol
Netware NLSP	Link-state protocol
OSPF	Link-state protocol
RIP v1	Distance vector protocol
RIP v2	Distance vector protocol

Advantages of Routing

- **Routing:** Helps send data efficiently by choosing the best path.
- **Automated Routing:** Uses algorithms to reduce manual work but gives less direct control.
- **Manual Routing:** Provides better control but requires more effort to manage.
- **Scalable Routing:** Works well for large networks and supports load balancing to reduce traffic and data loss.

Disadvantages of Routing

- **Static Routing:** Suitable only for small networks; difficult to manage and less secure in large networks.
- **Dynamic Routing:** Offers less manual control and uses more bandwidth and system resources.
- **Default Routing:** Easy to use but can cause issues if the default path is not properly configured.

Routing Metrics

The cost between nodes is measured using different routing metrics.

- **Hop Count:** The number of nodes a data packet passes through to reach its destination. Fewer hops mean a shorter and faster path.

- **Bandwidth Consumption:** Routing selects paths that use network bandwidth efficiently to avoid congestion.

- **Delay:** The time taken by a data packet to travel from source to destination. Routing aims to reduce delays.

- **Load:** Refers to how busy a network path is. Packets are sent through less busy paths for faster delivery.

- **Reliability:** Routing chooses stable and available paths to ensure data reaches its destination safely.

Convergence Issues

- Factors affecting the convergence time include the following:
  - **Routing protocol used**
  - **Distance of the router, or the number of hops from the point of change**
  - **Number of routers in the network that use dynamic routing protocols**
  - **Bandwidth and traffic load on communications links**
  - **Load on the router**
  - **Traffic patterns in relation to the topology change**

Route Calculation Fundamentals

- The capability of a routing protocol to update and calculate routes efficiently is based on several factors:

  - **Whether the protocol calculates and stores multiple routes to each destination.**

  - **The manner in which routing updates are initiated.**

  - **The metrics used to calculate distances or costs.**

The Initiation of Routing Updates

- **Routing protocols can instruct a router to update neighbors after a specific amount of time has passed, after a certain event has occurred, or both.**
- **Time-driven routing protocols wait for the update timer to expire and then send an update.**
- **Protocols that are event-driven do not require the router to update neighbors until the router detects a change in the network topology.**
- **Routing protocols that are exclusively time-driven react poorly to topology changes.**
- **Routing protocols that are exclusively event-driven could go for extended periods of time without sending updates.**

LAB

- Configuration of Dyanamic Routing-RIP

Introduction to EIGRP

EIGRP

- Improved Interior Gateway Routing Protocol (IGRP)
- Hybrid
  - More like distance vector, but with some link state ideas
- Scalable
- Support for Classless inter-Domain Routing (CIDR) and Variable Length subnet Mask(VLSM)
- Multi-protocol support
  - PDMs (protocol dependant modules)
- Rapid convergence with DUAL
- Partial, bounded updates

EIGRP Features

- **Cisco proprietary and is Hybrid:** (**containing Advanced distance vector and several link-state features**)
- **Fast convergence : stores all its neighbors' routing tables so that it can quickly adapt to alternate routes**. If no appropriate route exists, EIGRP queries its neighbors to discover an alternate route which continues until an alternate route is found.
- **Support for VLSM and discontiguous subnets:** as a classless routing protocol, **EIGRP advertises a subnet mask for each destination network; enabling it to support dis contiguous subnet works and VLSM**. Routes are automatically summarized at the major network number boundary, and can be configured to summarize on any bit boundary on any router interface.

Cont'd

- **Partial updates:** **Sends partial triggered updates as opposed to periodic updates--sent only when the path or metric changes** containing information about the changed route. For this reason EIGRP consumes significantly less bandwidth than IGRP. This behavior is different than that of link-state protocols, in which an update is transmitted to *all* link-state routers within an area.
- **Support for multiple network-layer protocols:** EIGRP supports IP, AppleTalk, and Novell NetWare Internet Packet Exchange (IPX**) through the use of protocol-dependent module**s. The rapid convergence and sophisticated metric structure of EIGRP offers superior performance and stability when implemented in IPX and AppleTalk networks.

Cont'd

- **Flexible network design**
- **Multicast and unicast instead of broadcast address:** EIGRP uses multicast and unicast, rather than broadcast. The multicast address used for EIGRP is 224.0.0.10.
- **Manual summarization at any point**
- **100% loop-free classless routing**
- **Easy configuration for WANs and LANs**
- **Load balancing**

EIGRP Key Technologies

- **Neighbor discover/recovery: Uses Hello packets between neighbors**
- **Reliable Transport Protocol (RTP): Guaranteed, ordered delivery of EIGRP packets to all neighbors**
- **DUAL finite-state machine: Selects lowest-cost, loop-free, paths to each destination**
- **Protocol-dependent modules (PDMs) : EIGRP supports IP, AppleTalk, and Novell NetWare; Each protocol has its own EIGRP module and operates independently from any of the others that may be running**

The Diffusing Update Algorithm (DUAL)

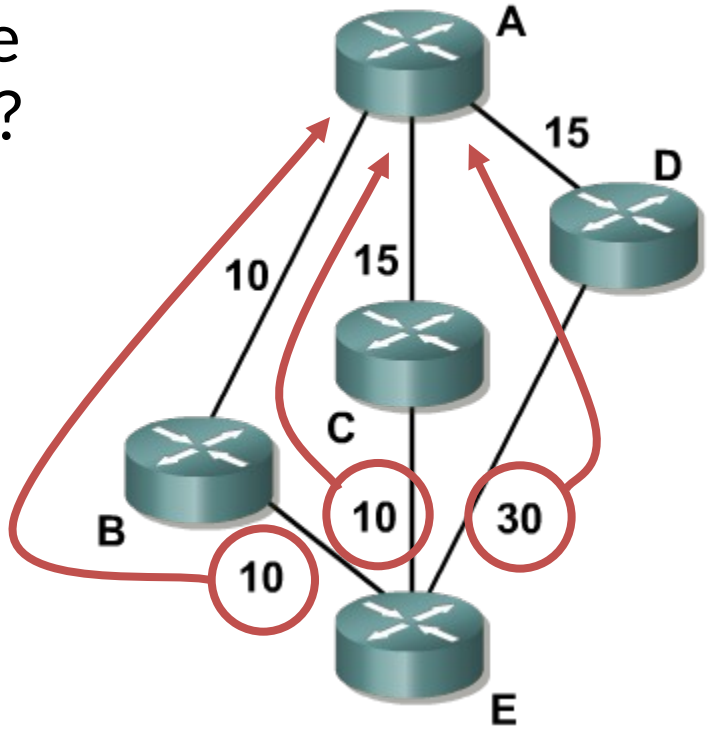
- How does EIGRP determine which routes are loop-free?
  - **Each of A's neighbors is reporting reachability to E**:

    B with a cost of 10

    C with a cost of 10

    D with a cost of 30

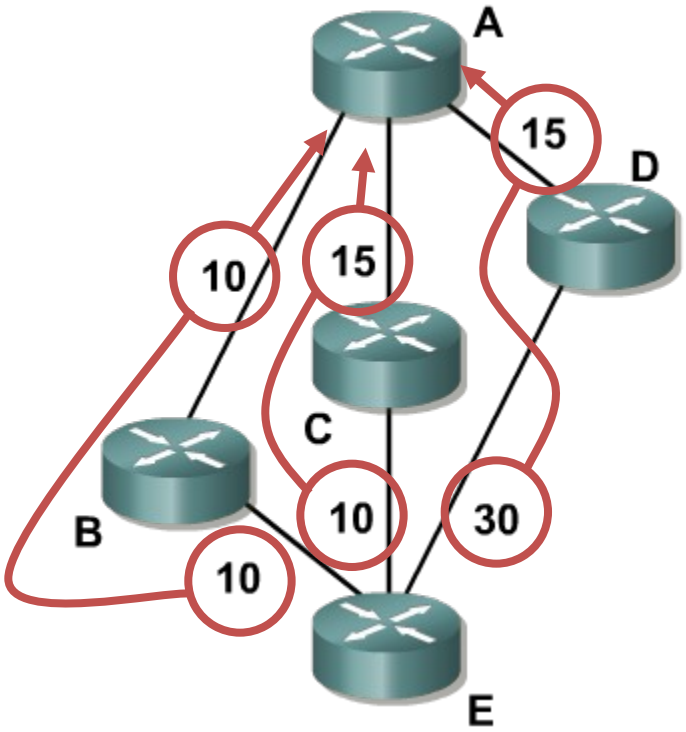
  - These three costs are called the reported distance (RD); **the distance each neighbor is reporting to a given destination**



- DUAL uses distance information (cost) to select efficient, loop-free paths.
- Lowest-cost route is calculated by adding the cost between the next-hop router and the destination--**Reported Distance (RD)**—**to the cost between the local router and the next-hop router**

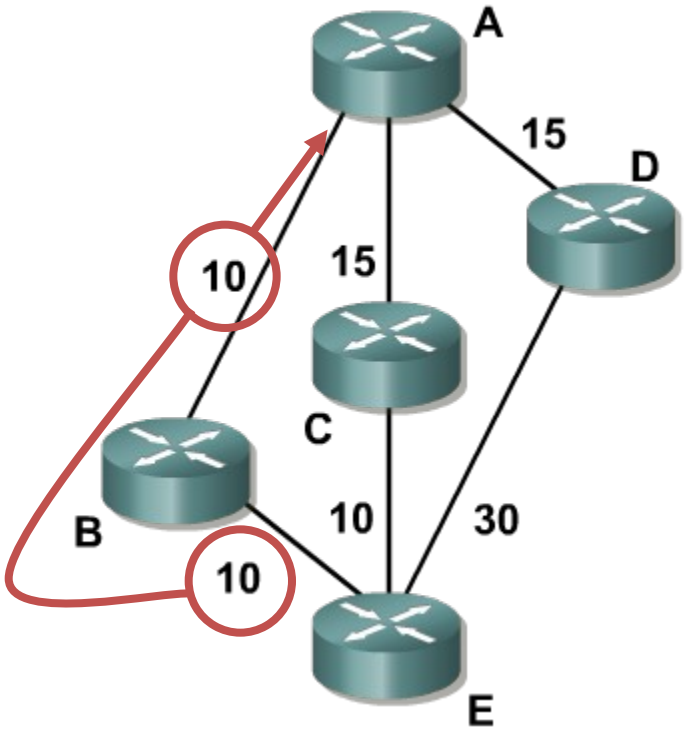
The Diffusing Update Algorithm (DUAL)

- At A, the total cost to reach E is:
  - 20 through B
  - 25 through C
  - 45 through D
- The best of these three paths is the path through B, with a cost of 20
- This is the feasible distance (FD)



The Diffusing Update Algorithm (DUAL)

- A uses the **FD** and the **RD** to determine which paths **are loop-free**
- The best path (FD) is used as a **benchmark**; all paths with RDs lower than the FD **cannot contain loops**
- The algorithm may mark some loop-free paths as loops
- However, it is guaranteed never to mark a looped path as loop-free



Successor (current successor): neighboring router that has the least-cost path to a destination (the lowest FD) guaranteed not to be a part of the routing loop (used for forwarding packets.
Multiple successors can exist if they have the same FD

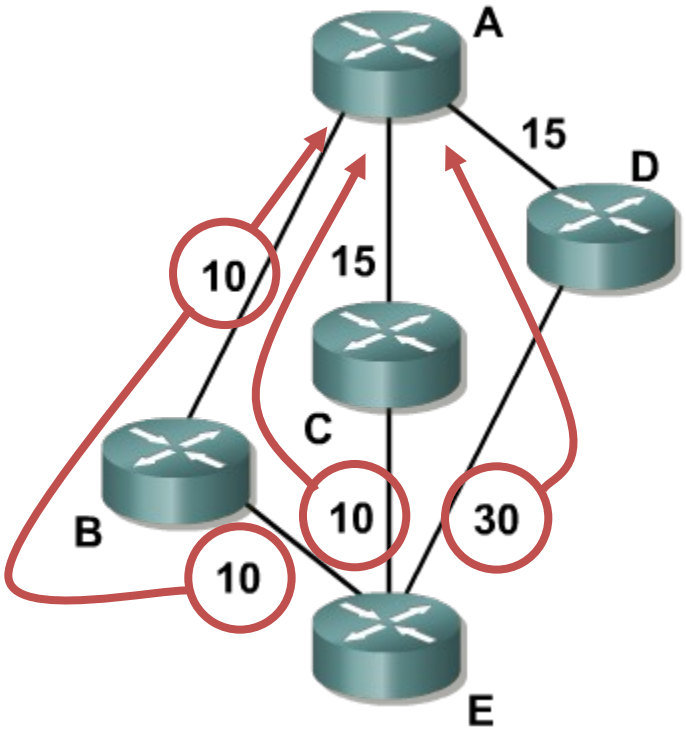
The Diffusing Update Algorithm (DUAL)

- At A:

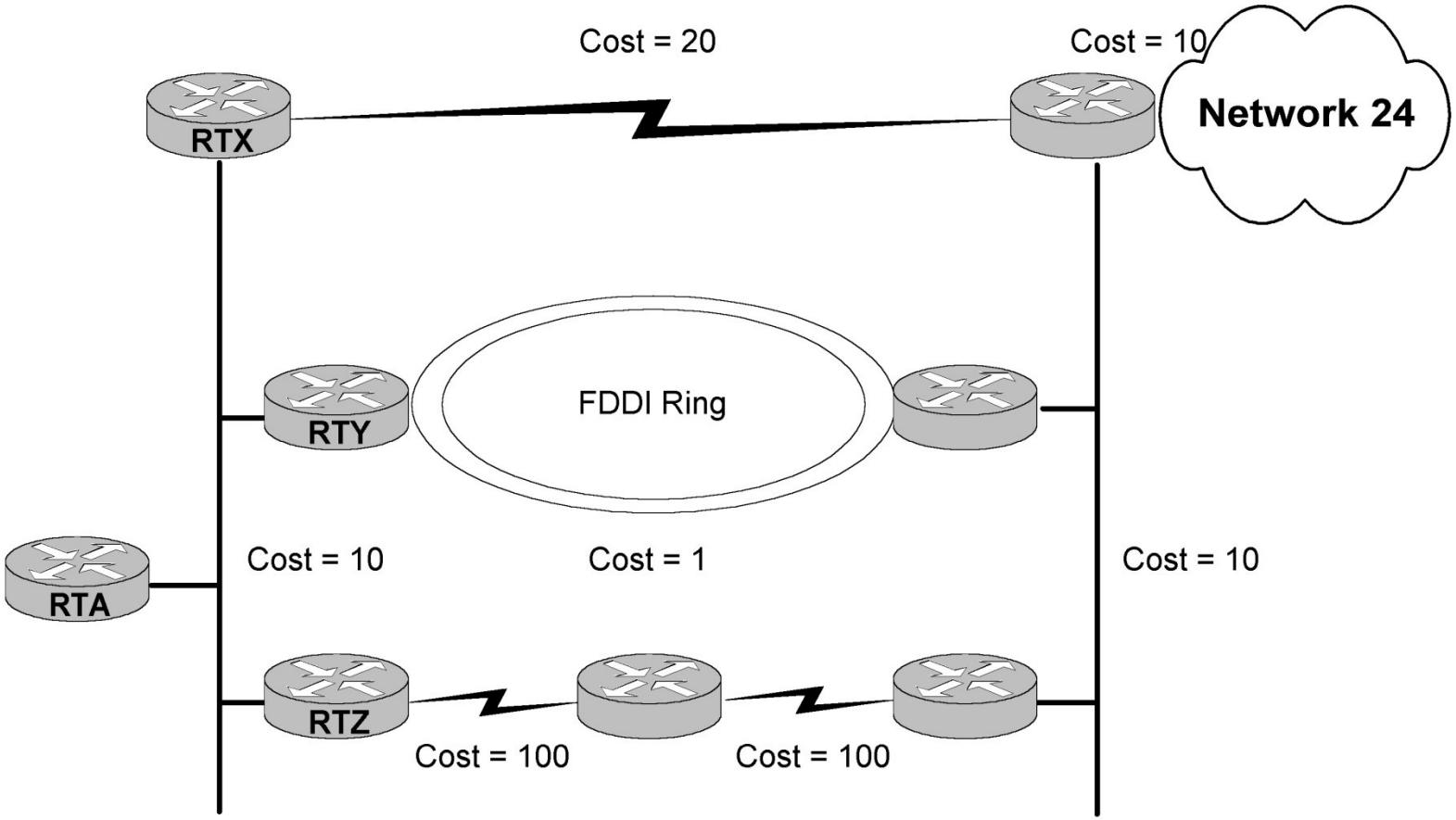
  The path through B is the best path (FD), at 20

  C can reach E with a cost of 10; **10 (RD) is less than 20 (FD), so this path is loop-free.**

  D can reach E with a cost of 30; **30 (RD) is not less than 20 (FD), so EIGRP assumes this path is a loop.**



DUAL


Cost = 20
Cost = 10
RTX
Network 24
FDDI Ring
RTY
RTA
Cost = 10
Cost = 1
Cost = 10
RTZ
Cost = 100
Cost = 100

DUAL

Neighbor	Feasible Distance to Net 24	Reported Distance to Net 24
RTY	31	21
RTZ	230	220
RTX	40	30

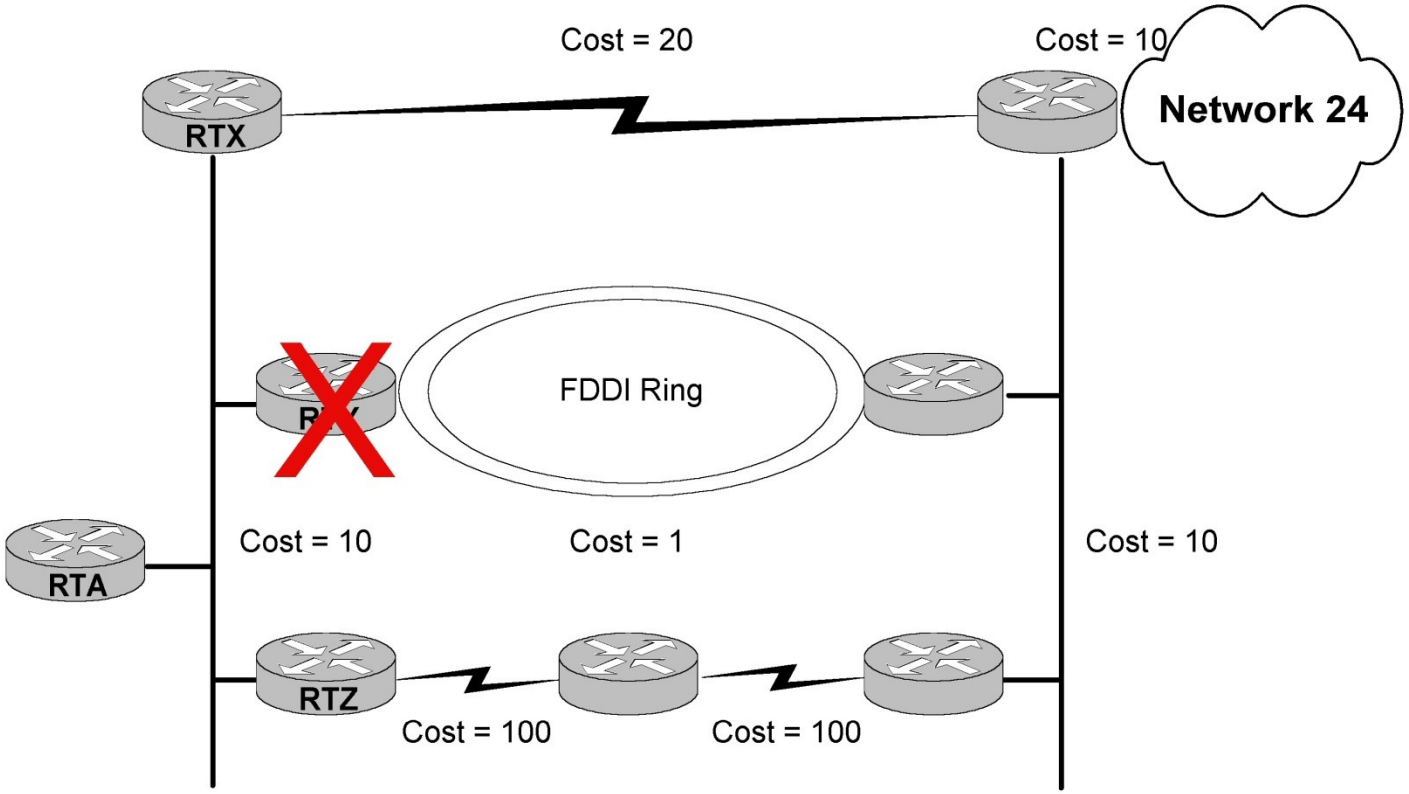
RTY is successor with a computed cost (feasible distance) of 31.

“31” is the Feasible Distance (FD).

RTX is a feasible successor **because its RD is less than the FD**.

RTZ is not a feasible successor because it’s **RD (230) > RTY’s FD (31)** and thus may be routed through RTY (loop ?)

DUAL


Cost = 20
Cost = 10
RTX
Network 24
FDDI Ring
Cost = 10
Cost = 1
Cost = 10
RTA
RTZ
Cost = 100
Cost = 100

DUAL

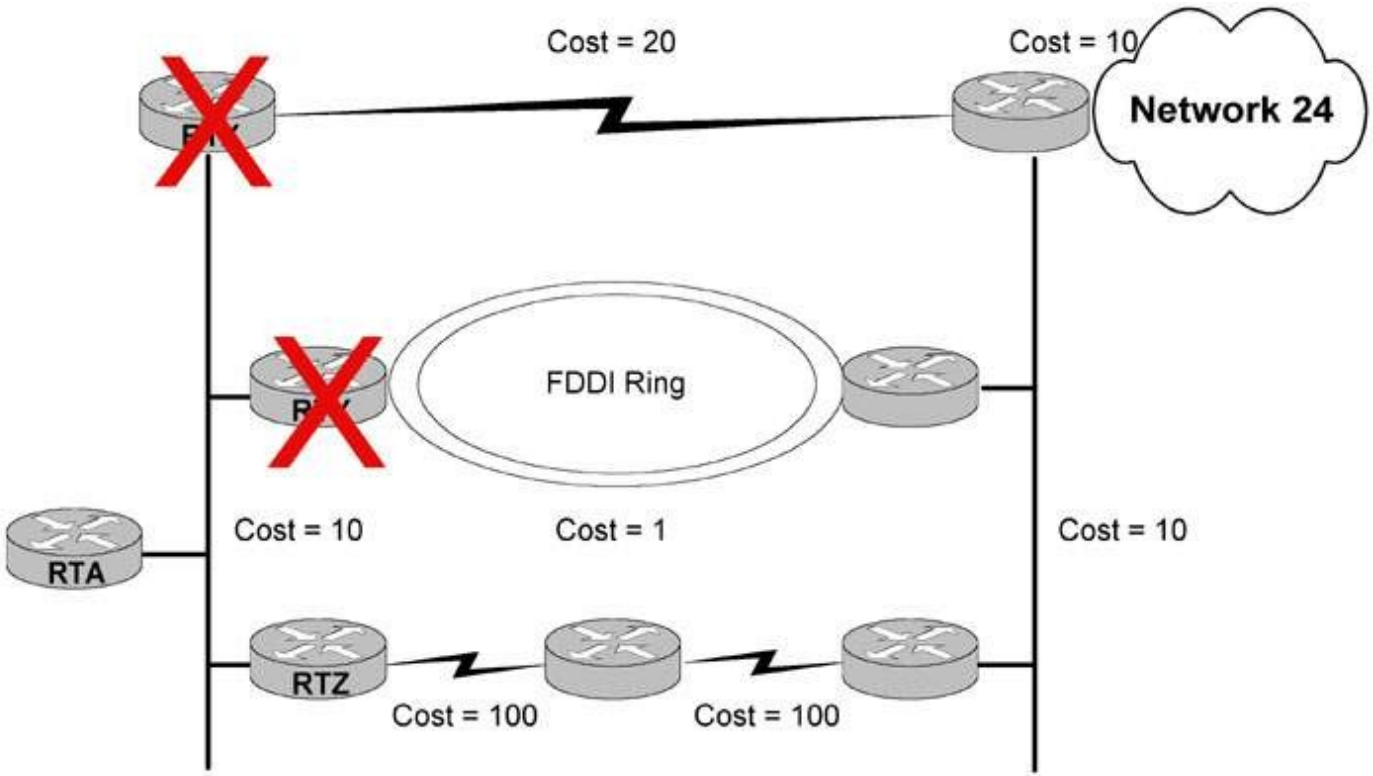
Neighbor	FD to Net 24	Reported Distance to Net 24
RTY	31	21
RTZ	230	220
RTX	40	30



Since RTX is a feasible successor, it is installed in the routing table immediately (no recomputation).

It's RD (30) is less than the FD (31).

DUAL


Cost = 20
Cost = 10
Network 24
FDDI Ring
Cost = 10
Cost = 1
Cost = 10
RTA
RTZ
Cost = 100
Cost = 100

DUAL

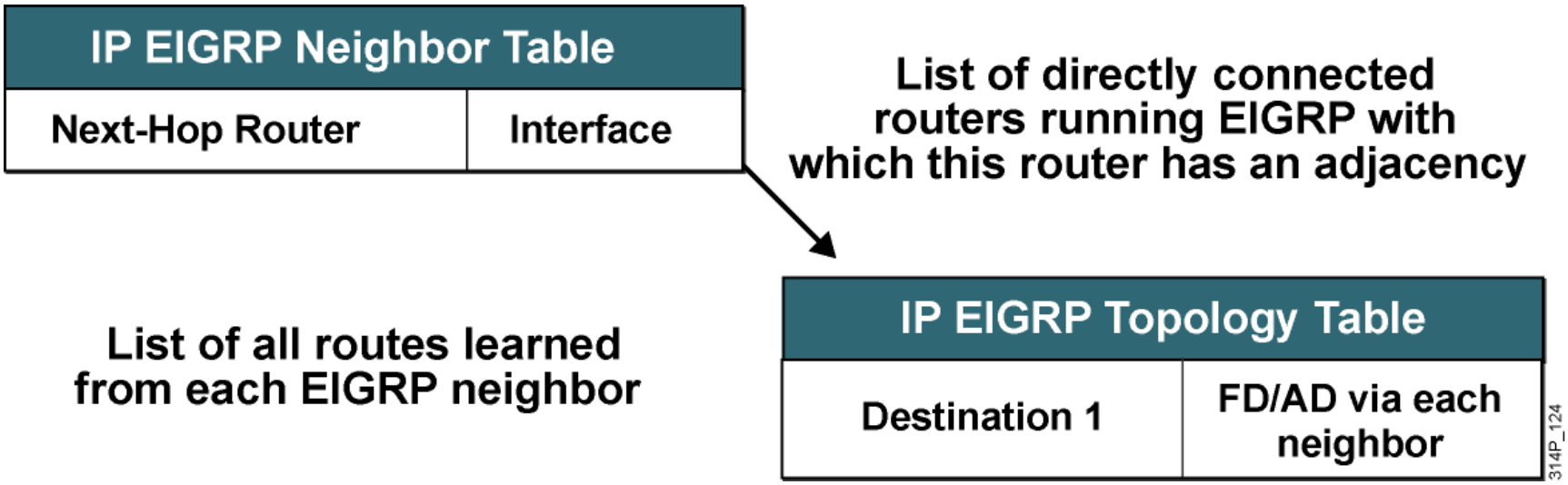
Neighbor	Computed Cost to Net 24	Reported Distance to Net 24
RTY	31	21
RTZ	230	220
RTX	40	30

RTY is not a feasible successor. It's RD (220) is greater than the FD (31) for Net 24. Before this route can be installed, the route to net 24 must be placed in the ***active state*** and recomputed.

EIGRP Topology Table

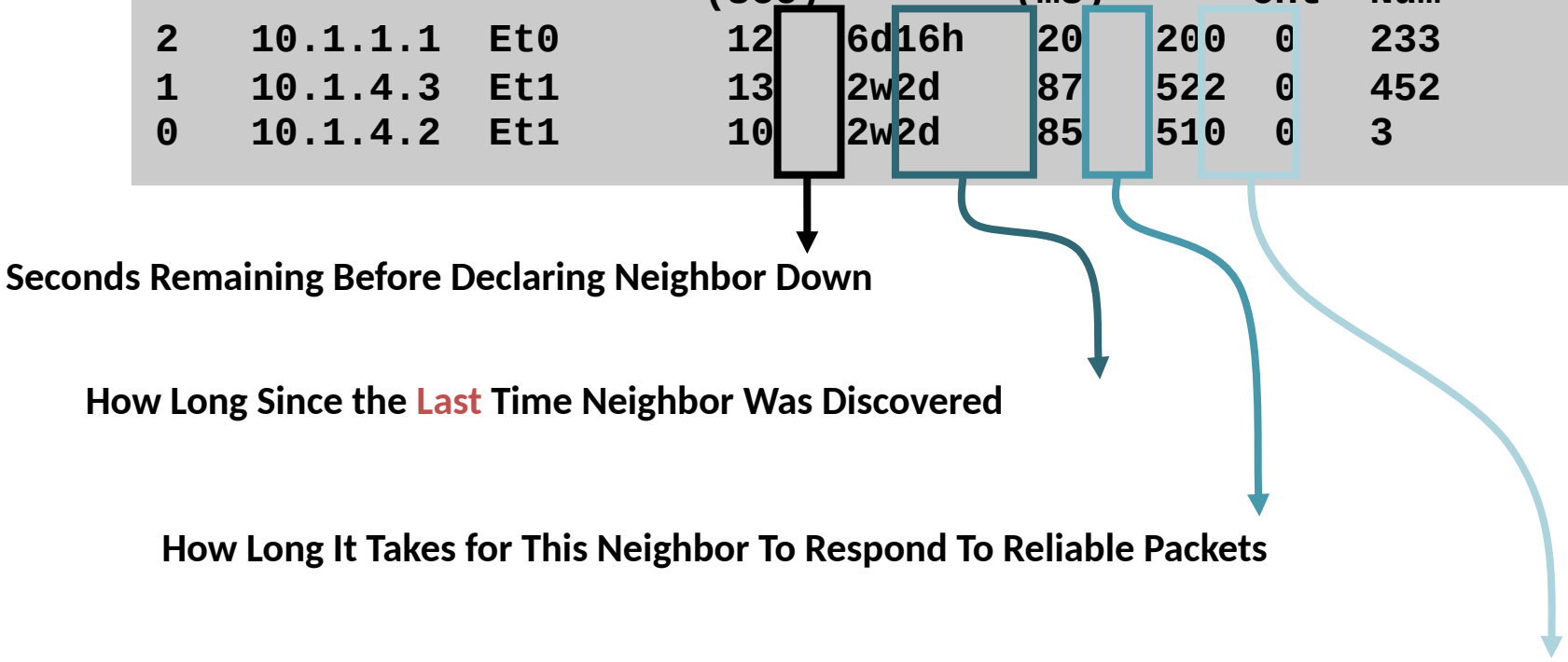
- When a **router discovers a new neighbor, an update is sent to and received from its new neighbor** populating the **topology table** (containing destinations advertised by all neighbors)
- The topology table:
  - **updated when a directly connected route or interface changes or when a neighboring router reports a change to a route**
  - **Entry for a destination exists in either active or passive state**:
    - **Passive state: router is not performing a recomputation**
    - **Active state: router is performing a recomputation**
  - **Recomputation occurs when the destination has no feasible successors** (initiated by sending a query packet to each of the neighboring routers

EIGRP Topology Table



EIGRP Neighbor Status

```
RTRA#show ip eigrp neighbors
IP-EIGRP neighbors for process 1
H   Address   Interface Hold Uptime SRTT  RTO  Q   Seq
                        (sec)       (ms)     Cnt  Num
2   10.1.1.1  Et0        12  6d16h  20   200  0   233
1   10.1.4.3  Et1        13  2w2d   87   522  0   452
0   10.1.4.2  Et1        10  2w2d   85   510  0   3
```



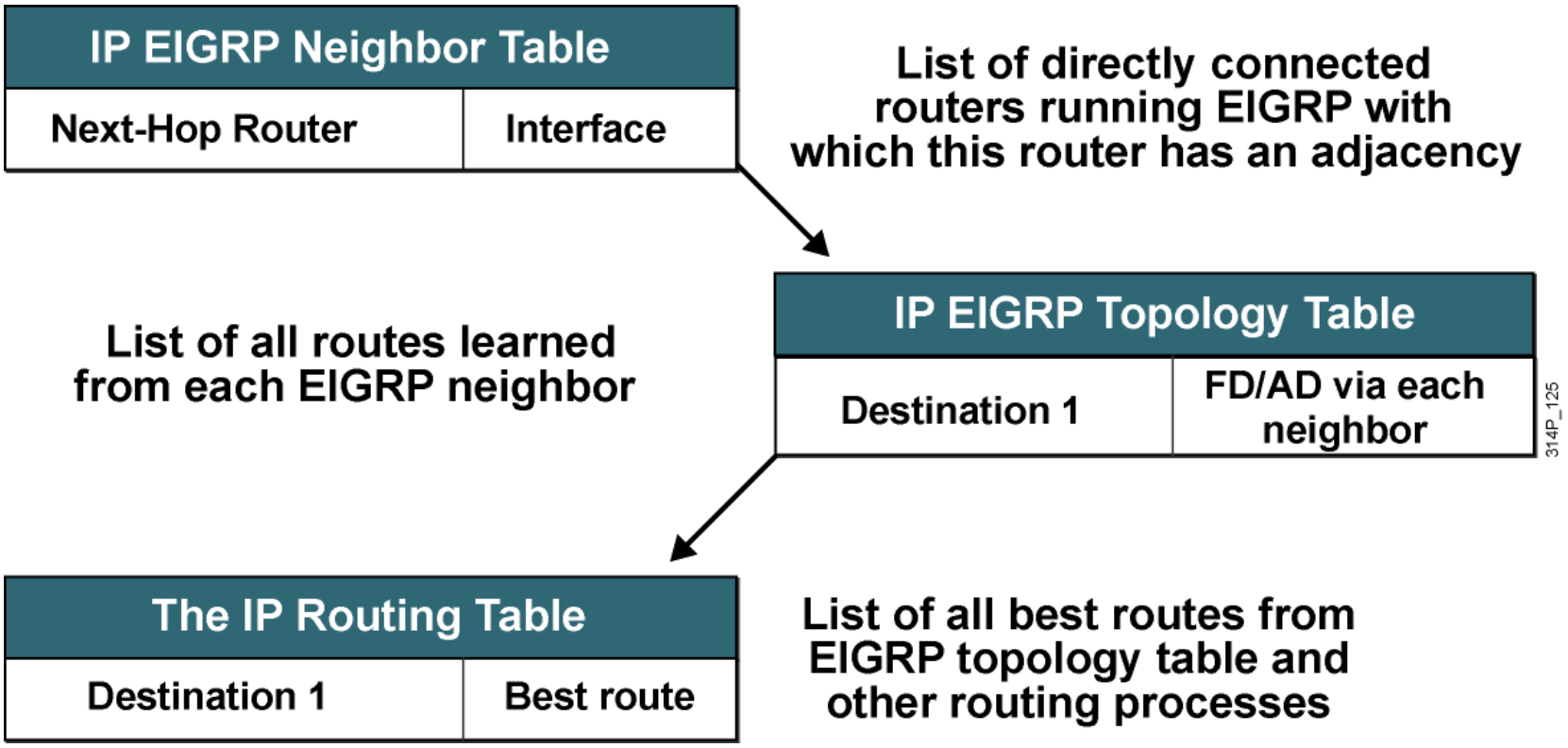
Seconds Remaining Before Declaring Neighbor Down

How Long Since the Last Time Neighbor Was Discovered

How Long It Takes for This Neighbor To Respond To Reliable Packets

How Long to Wait Before Retransmitting If No Acknowledgement

EIGRP IP Routing Table



Example: EIGRP Tables

Router C's tables:

IP EIGRP Neighbor Table

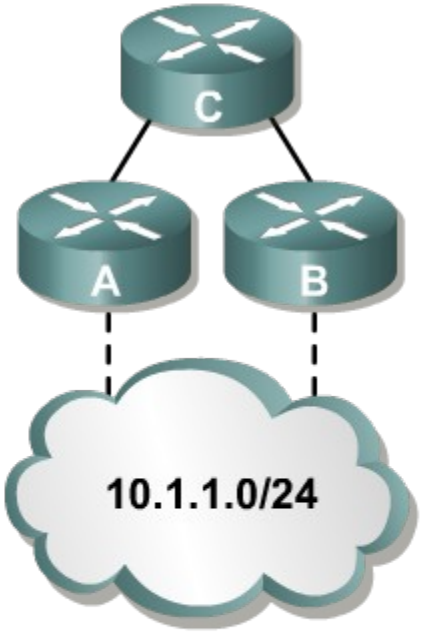
Next-Hop Router	Interface
Router A	Ethernet 0
Router B	Ethernet 1

IP EIGRP Topology Table

Network	Feasible Distance (EIGRP Metric)	Advertised Distance	EIGRP Neighbor
10.1.1.0 /24	2000	1000	Router A (E0)
10.1.1.0 /24	2500	1500	Router B (E1)

The IP Routing Table

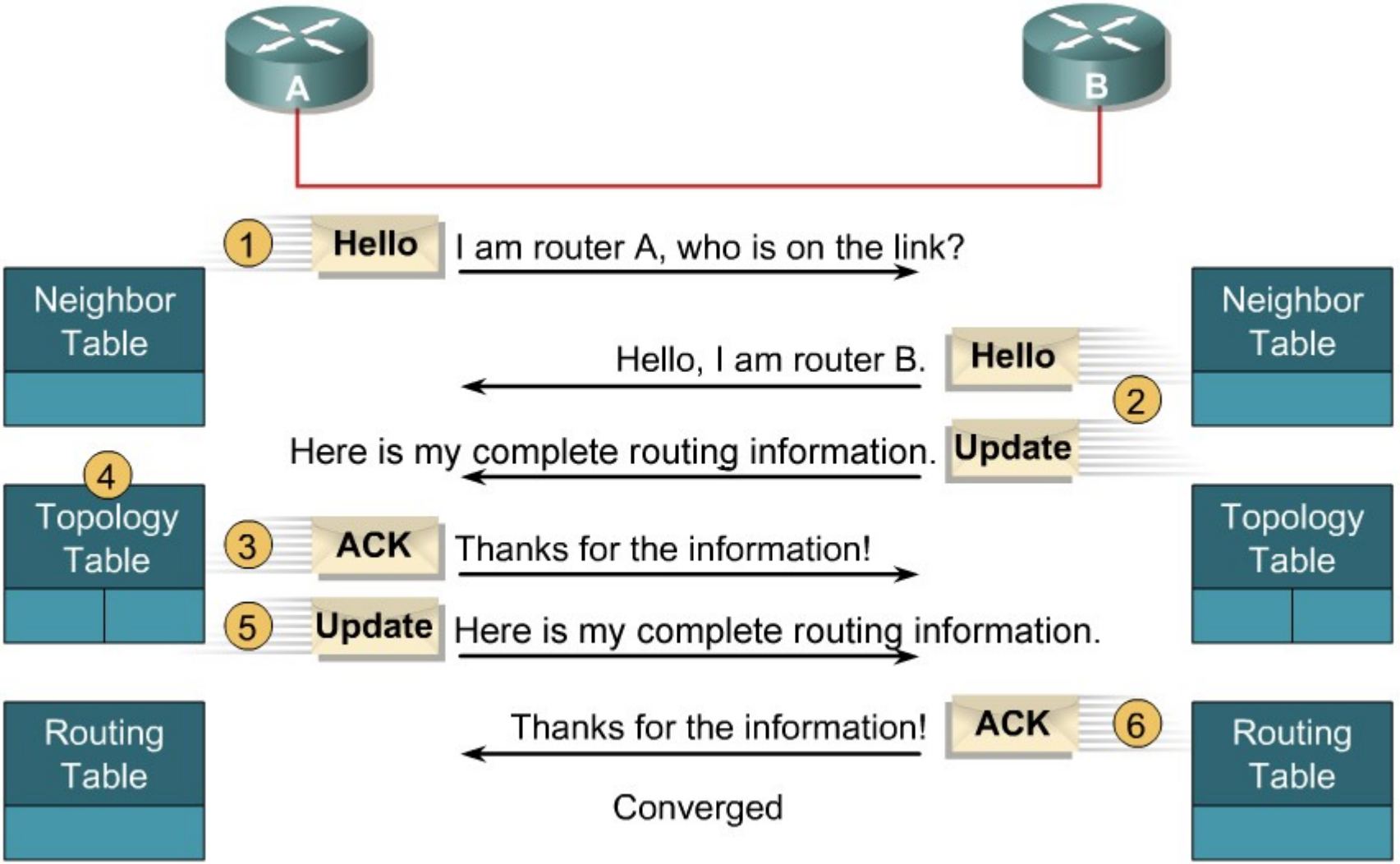
Network	Metric (Feasible Distance)	Outbound Interface	Next Hop (EIGRP Neighbor)
10.1.1.0 /24	2000	Ethernet 0	Router A



EIGRP Packets

- **Hello:** Establish neighbor relationships.
- **Update:** Send routing updates
- Qu**ery:** Ask neighbors about routing information
- **Reply:** Respond to query about routing information
- **ACK:** Acknowledge a reliable packet

Initial Route Discovery



EIGRP Metric

- Same metric components as IGRP:
  - Bandwidth
  - Delay
  - Reliability
  - Loading
  - MTU (Maximum Transmission Unit)
- EIGRP metric is IGRP metric multiplied by 256

EIGRP Metric

This metric can be based on five criteria, but EIGRP uses only two of these criteria by default:

- **Bandwidth:** The smallest bandwidth between source and destination
- **Delay:** The cumulative interface delay along the path

Other criteria can be used, but are not recommended, because they typically result in frequent recalculation of the topology table:

- **Reliability:** This value represents the worst reliability between source and destination, based on keepalives.
- **Loading:** This value represents the worst load on a link between source and destination, computed based on the packet rate and the configured bandwidth of the interface.
- **MTU:** This criterion represents the smallest MTU in path. MTU is included in the EIGRP routing update but is not actually used in the metric calculation.

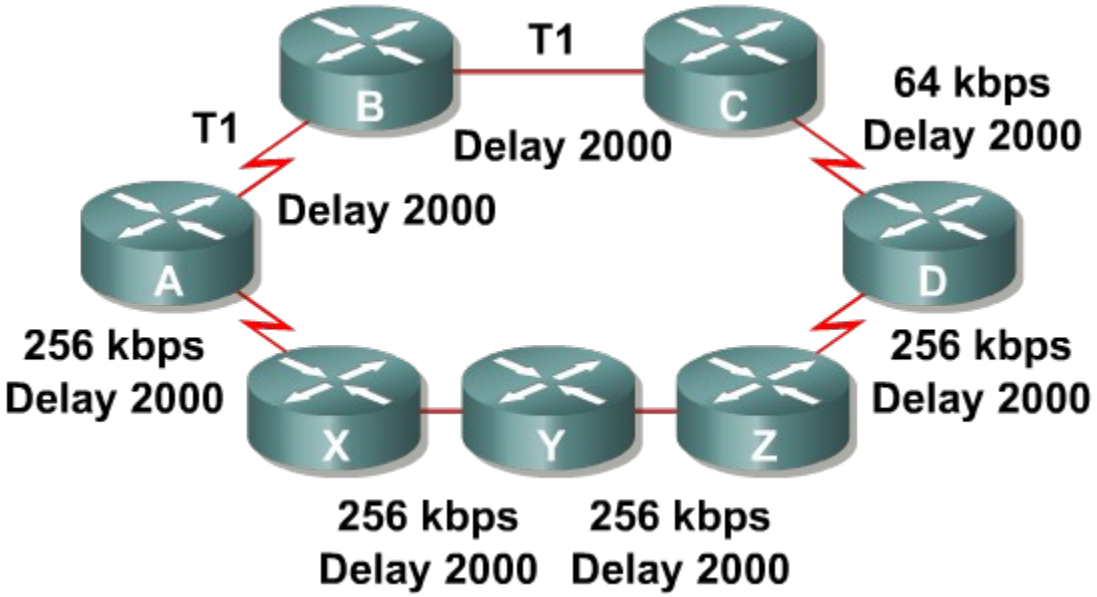
EIGRP Metric Calculation

- By **default**, EIGRP metric:

  Metric = bandwidth (slowest link) + delay (sum of delays)

- **Delay** = sum of the delays in the path, in tens of microseconds, multiplied by 256.
- **Bandwidth** = [10^7 / (minimum bandwidth link along the path, in kilobits per second)] * 256

EIGRP Metrics Calculation Example



A → B → C → D **Least bandwidth 64 kbps** **Total delay 6,000**

A → X → Y → Z → D **Least bandwidth 256 kbps** **Total delay 8,000**

- Delay is the sum of all the delays of the links along the paths:
  Delay = [delay in tens of microseconds] x 256
- BW is the lowest bandwidth of the links along the paths:
  BW = [10,000,000 / (bandwidth in kbps)] x 256

LAB

- Configuration of Dyanamic Routing-EIGRP

OSPF

Characteristics of OSPF's hierarchical design

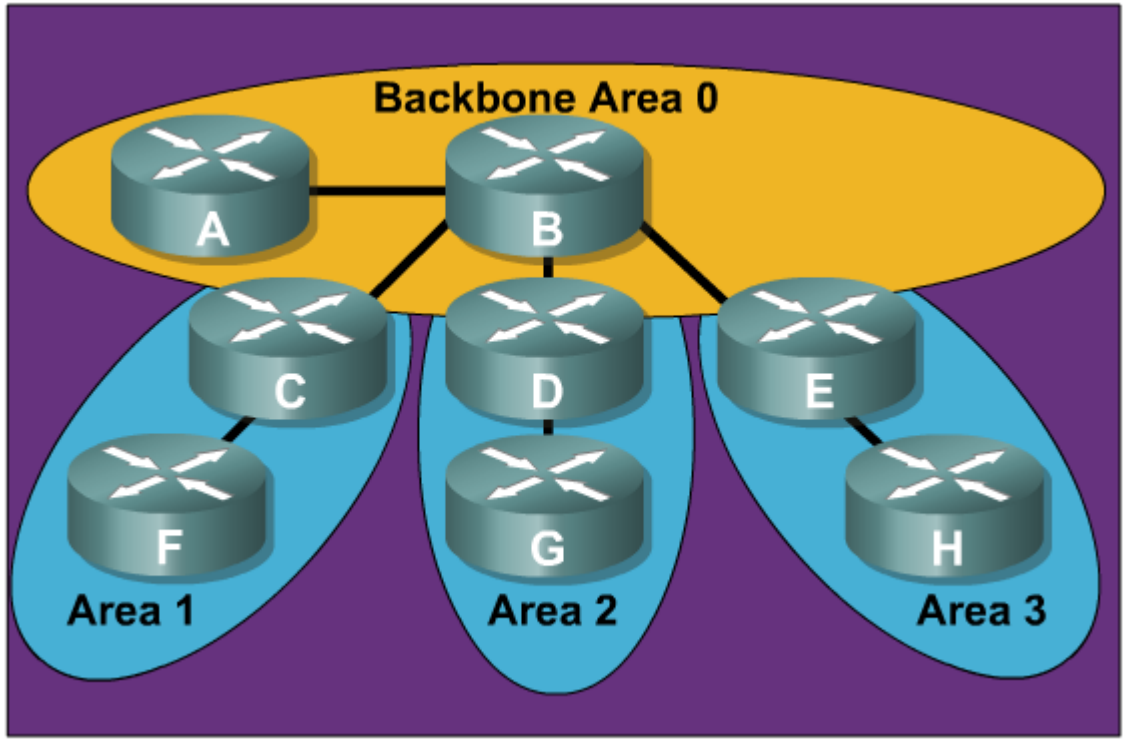
First, let's start with a review of the characteristics of OSPF's hierarchical design. Recall that **OSPF allows for the creation of multiple areas so that the network administrator can**:

- **Reduce the size of routing tables**
- **Isolate topology changes as much as possible to the area in which they occur**
- **Allow only summary Link state advertisements (LSA) updates to cross area boundaries and Reap all the benefits of using a hierarchical addressing scheme**.

OSPF Areas

Review of OSPF area characteristics:

- **Minimizes routing table entries**
- **Localizes impact of a topology change within an area**
- **Detailed Link State Advertisement (LSA) flooding stops at the area boundary**
- **Requires a hierarchical network design**



OSPF Areas

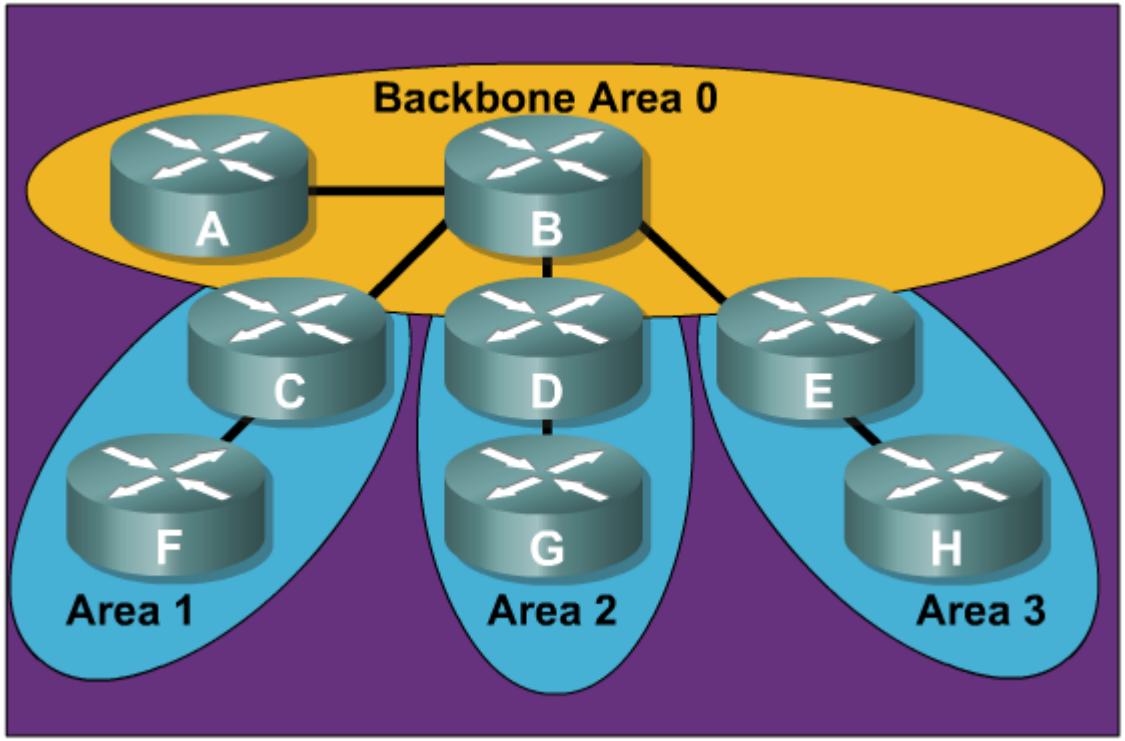
New terminology for areas:

- **Transit Area:**

  Also known as **Backbone Area 0**

- **Regular Area:**

  Also known as **Nonbackbone areas**



OSPF Areas

- The **backbone area** is also referred to as **a transit area**.
- Another type of **transit area** is one that is configured with **virtual links**.
- **Virtual links** are a temporary solution when an organization has two **backbones that are physically and logically disconnected**.
- A virtual link between the two backbones is also called a **transit area**.
- All other areas are known as **regular areas** (**standard areas, stub areas, totally stubby areas, and NSSAs)**

OSPF Database

- OSPF maintains three databases
- Adjacency Database (**`show ip ospf neighbor`**)
- Link-state Database (**`show ip ospf database`**)
- Forwarding Database (**`show ip route`**)

OSPF Database

- Remember that OSPF uses three databases in its operation:

- The **Adjacency Database** is a table listing all **OSPF neighbors with which the local router has established bidirectional communication.** Display this table with the **show ip ospf neighbor** command.

- The **Link-state Database** is a table listing all **other routers in the network. This database shows the network topology**. **All OSPF routers in the same area have identical Link-state Databases.** Display this table with the **show ip ospf database** command.

- **The Forwarding Database** is **simply the routing table**. It includes a list of all routes generated when the SPF algorithm is calculated on the Link-state Database. Display this table with the **show ip route** command.

What is LSDB?

- LSDB is an acronym for **L**ink-**s**tate **D**ata**b**ase.

- LSAs and the Link-state Database

LSA Sequence Numbering

- Each LSA in the LSDB **maintains a sequence number**.
- The sequence numbering scheme is a **4-byte number** that begins with **0x80000001 and ends with 0x7…..F**.
- OSPF floods each **LSA every 30 minutes** to **maintain proper database synchronization**. **Each time the LSA is flooded, the sequence number is incremented by one**.
- Ultimately, an **LSA sequence number will wrap around to 0x80000001.** When this occurs, **the existing LSA is prematurely aged to maxage (one hour) and flushed**.
- When a **router encounters two instances of an LSA**, **it must determine which is more recent.** The **LSA having the newer (higher) LS sequence number is more recent**.

The OSPF Link-State Database

- The **`show ip ospf database`** command displays the current LSDB for the local router.

```
RTC#show ip ospf database

            OSPF Router with ID (192.168.1.253) (Process ID 3)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
192.168.1.249   192.168.1.249   1705        0x80000005 0x00D5B0 5
192.168.1.253   192.168.1.253   1578        0x80000006 0x009F91 5
```

Verifying LSA Age and Sequence Number

- In the first output below, notice the age timer will expire sometime after 1800 seconds or 30 minutes.

```
RTC#show ip ospf database

            OSPF Router with ID (192.168.1.253) (Process ID 3)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
192.168.1.249   192.168.1.249   1705        0x80000005 0x00D5B0 5
192.168.1.253   192.168.1.253   1578        0x80000006 0x009F91 5
```

- A few minutes later, the router has received an LSU for both links. Note the refreshed age timer and incremented sequence number.

```
RTC#show ip ospf database

            OSPF Router with ID (192.168.1.253) (Process ID 3)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
192.168.1.249   192.168.1.249   106         0x80000006 0x00D3B1 5
192.168.1.253   192.168.1.253   58          0x80000007 0x009D92 5
```

Basic OSPF Configuration

Configuring OSPF

- An excellent resource for information on the many different OSPF configurations is the Cisco white paper, “Configuring OSPF”, which can be downloaded from the Cisco website:

  http://www.cisco.com/en/US/products/sw/iosswrel/ps1835/products_configuration_guide_chapter09186a00800b3f2e.html

Configuring Basic OSPF

Router(config)#

router ospf *process-id* **[vrf** *vpn-name***]**

- Enable one or more OSPF routing processes.

Router(config-router)#

network *ip-address wildcard-mask* **area** *area-id*

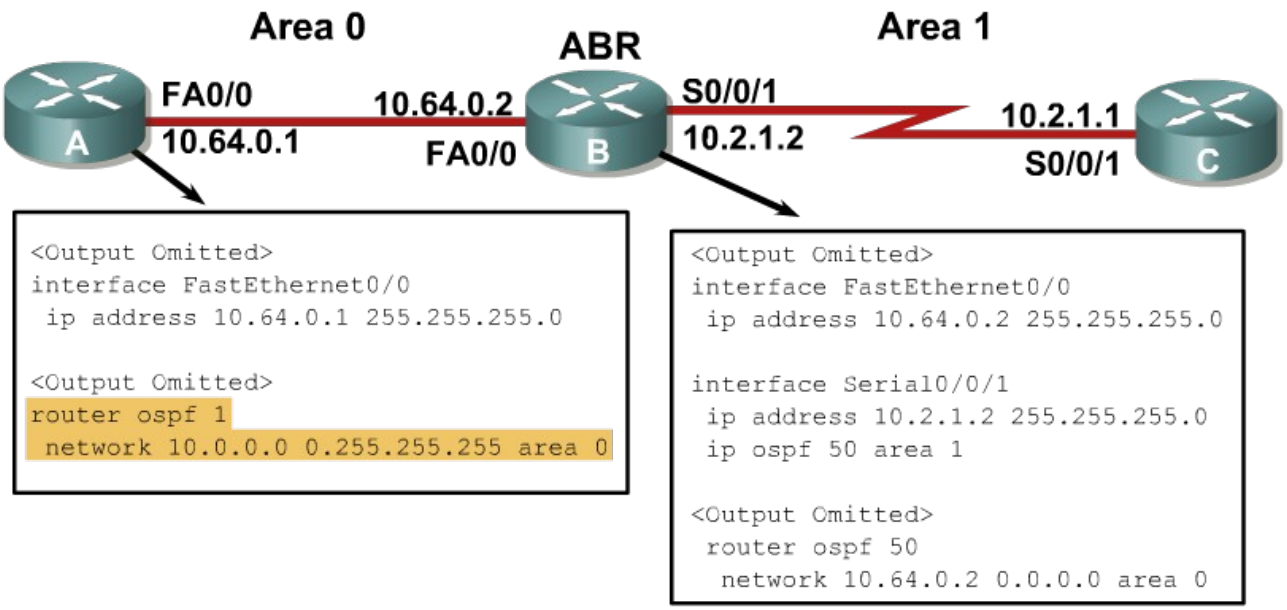
- Define the interfaces that OSPF will run on.

Router(config-if)#

ip ospf *process-id* **area** *area-id* **[secondaries none]**

- Optional method to enable OSPF explicitly on an interface.

Configuring OSPF for Multiple Areas



Example: Configuring OSPF for Multiple Areas

The figure shows an example of multiarea OSPF configuration. Router A is in area 0, router C is in area 1, and router B is the area border router (ABR) between the two areas.

The configuration for router A is nothing new. However although Router B has a **network** statement for area 0 within the OSPF routing process, notice that the configuration for area 1 uses the **ip ospf 50 area 1** command on the interface.

Alternatively, we could have used a separate **network** router configuration command in the OSPF routing process.

OSPF Router ID

- The **router is known to OSPF by the OSPF router ID number**
- LSDBs use the **OSPF router ID to differentiate one router from the next.**
- By default, the **router ID is the highest IP address on an active interface at the moment of OSPF process startup**.

  If no interface is up when the OSPF process starts, you will get the following error message:

```
p5r2(config)#router ospf 1
2w1d: %OSPF-4-NORTRID: OSPF process 1 cannot start.
```

- A **loopback interface can override the OSPF router ID**. **If a loopback interface exists, the router ID is the highest IP address on any active loopback interface.**
- The OSPF `router-id` command can be used to override the OSPF router ID.
- Using a loopback interface or a `router-id` command is recommended for stability.

OSPF `router-id` Command

Router(config-router)# **`router-id`** *`ip-address`*

- This command is configured under the `router ospf [process-id]` command.
- Any unique arbitrary 32-bit value in an IP address format (dotted decimal) can be used.
- If this command is used on an OSPF process that is already active, then the new router ID takes effect after the next reload or after a manual restarting of the OSPF process using:

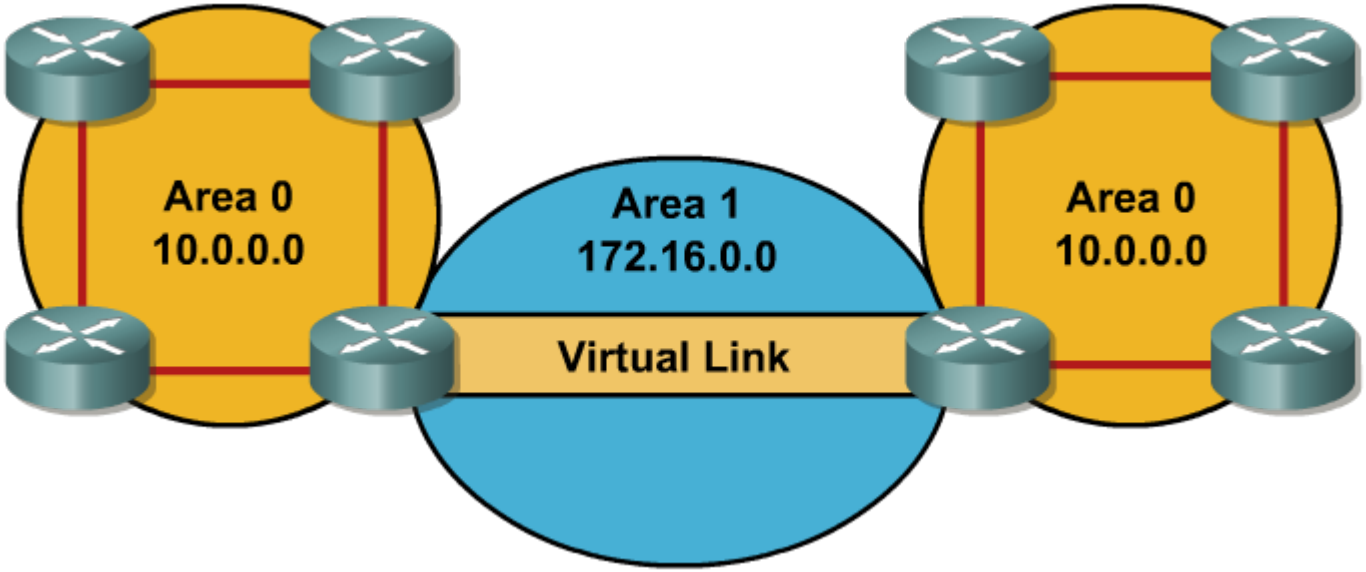
```
Router#clear ip ospf process
```

```
Router(config)#router ospf 1
Router(config-router)#router-id 172.16.1.1

Router#clear ip ospf process
```

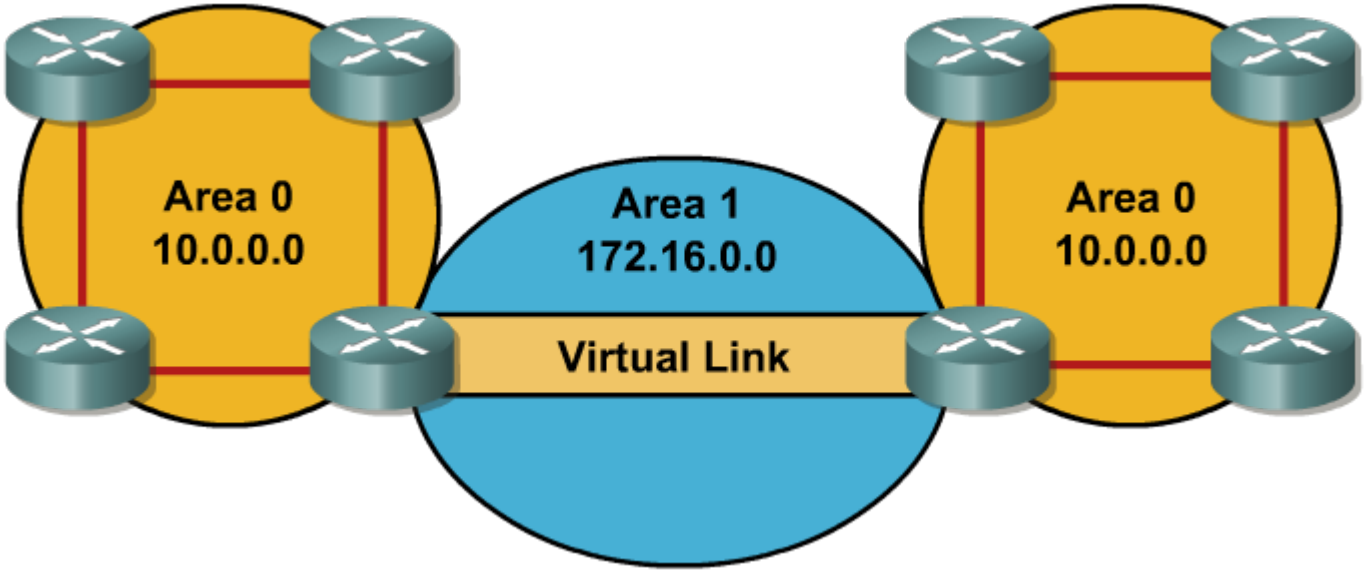
Virtual Link Concepts and Configuration

Virtual Links



- **Virtual links are used to connect a discontiguous area to area 0.**
- **A logical connection is built between router A and router B.**
- **Virtual links are recommended for backup or temporary connections.**

Special Treatment for LSAs on Virtual Links



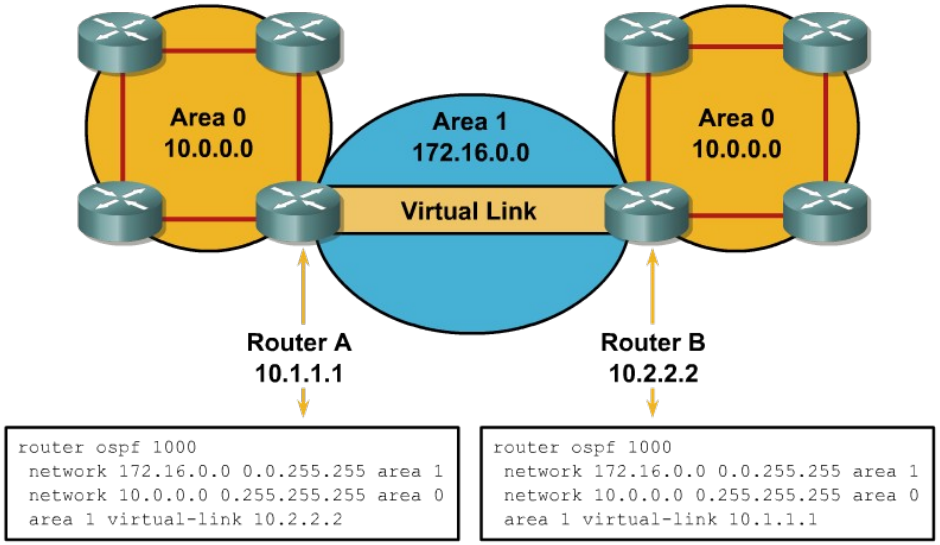
- **LSAs usually age out after 30 minutes**
- LSAs learned across virtual links have the **DoNotAge (DNA) option set**
- **Required to prevent excessive flooding over virtual links**

Configuring Virtual Links

```
Router(config-router)#
area area-id virtual-link router-id [authentication
[message-digest | null]] [hello-interval seconds]
[retransmit-interval seconds] [transmit-delay seconds]
[dead-interval seconds] [[authentication-key key] |
[message-digest-key key-id md5 key]]
```

- Creates a virtual link
- Optional parameters are used in place of **`interface ip ospf`** commands

Configuring and Verifying a Virtual Link




```
RouterA#sh ip ospf virtual-links
Virtual Link OSPF_VL0 to router 10.2.2.2 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 1, via interface Serial0/0/1, Cost of using 781
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
    Adjacency State FULL (Hello suppressed)
    Index 1/2, retransmission queue length 0, number of retransmission 1
    First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
    Last retransmission scan length is 1, maximum is 1
    Last retransmission scan time is 0 msec, maximum is 0 msec
```

OSPF Authentication

OSPF Authentication Types

- OSPF supports 2 types of authentication:
  - **Simple password authentication (plain text)**
  - **MD5 authentication**
- Router generates and checks each packet and authenticates the source of each update packet it receives
- Configure **a "key" (password)**
  - Note: **all participating neighbors must have the same key configured**

Configuring Simple Password Authentication

```
Router(config-if)#
ip ospf authentication-key password
```

- Assign a password to be used with neighboring routers.

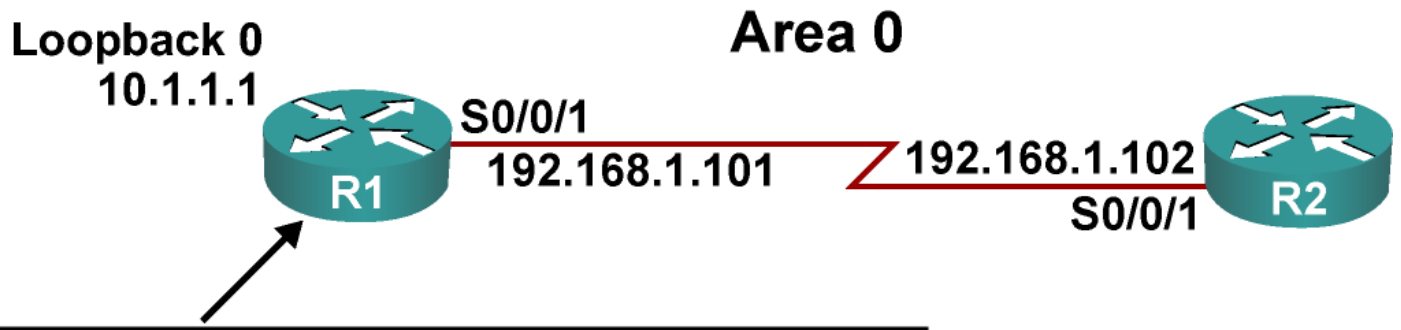
```
Router(config-if)#
ip ospf authentication [message-digest | null]
```

- Specifies the authentication type for an interface (since IOS 12.0).

```
Router(config-router)#
area area-id authentication [message-digest]
```

- Specifies the authentication type for an area (was in IOS before 12.0).

Example Simple Password Authentication Configuration




```
<output omitted>
interface Loopback0
 ip address 10.1.1.1. 255.255.255.0

<output omitted>
interface Serial0/0/1
 ip address 192.168.1.101 255.255.255.224
 ip ospf authentication
 ip ospf authetication-key plainpas

<output omitted>
router ospf 10
 log-adjacency-changes
 network 10.1.1.1 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

314P_076

R2 Configuration for Simple Password Authentication

```
<output omitted>
interface Loopback0
 ip address 10.2.2.2 255.255.255.0

<output omitted>
interface Serial0/0/1
 ip address 192.168.1.102 255.255.255.224
 ip ospf authentication
 ip ospf authentication-key plainpas

<output omitted>
router ospf 10
 log-adjacency-changes
 network 10.2.2.2 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

Verifying Simple Password Authentication

```
R1#sh ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:32    192.168.1.102   Serial0/0/1

R1#show ip route
<output omitted>
Gateway of last resort is not set
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
O       10.2.2.2/32 [110/782] via 192.168.1.102, 00:01:17, Serial0/0/1
C       10.1.1.0/24 is directly connected, Loopback0
     192.168.1.0/27 is subnetted, 1 subnets
C       192.168.1.96 is directly connected, Serial0/0/1

R1#ping 10.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/29/32 ms
```

Configuring OSPF MD5 Authentication

```
Router(config-if)#
ip ospf message-digest-key key-id md5 key
```

- Assign a key ID and key to be used with neighboring routers.

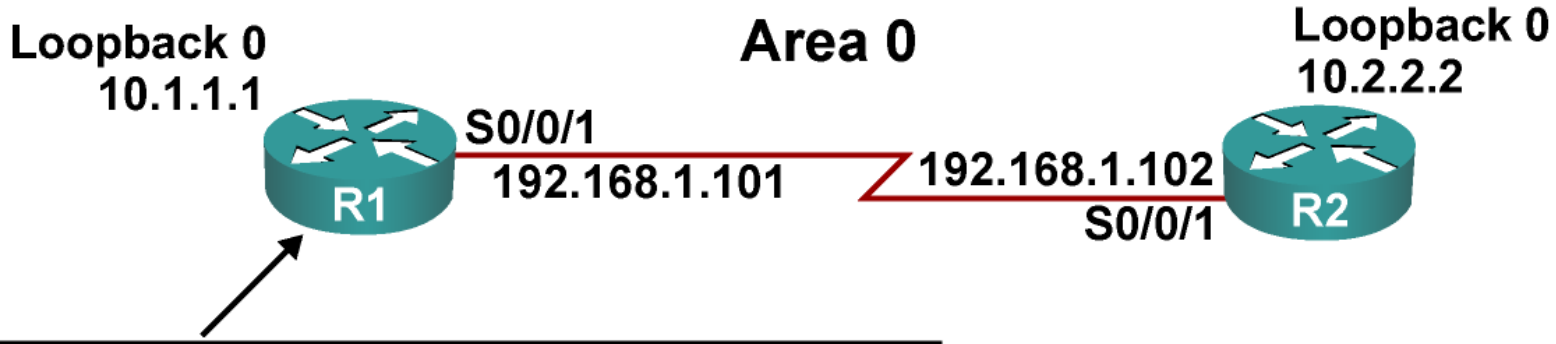
```
Router(config-if)#
ip ospf authentication [message-digest | null]
```

- Specifies the authentication type for an interface (since IOS 12.0).

```
Router(config-router)#
area area-id authentication [message-digest]
```

- Specifies the authentication type for an area (was in IOS before 12.0).

Example MD5 Authentication Configuration




```
<output omitted>
interface Loopback0
 ip address 10.1.1.1 255.255.255.0

<output omitted>
interface Serial0/0/1
 ip address 192.168.1.101 255.255.255.224
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 secretpass

<output omitted>
router ospf 10
 log-adjacency-changes
 network 10.1.1.1 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

314P_077

R2 Configuration for MD5 Authenticaiton

```
<output omitted>
interface Loopback0
 ip address 10.2.2.2 255.255.255.0

<output omitted>
interface Serial0/0/1
 ip address 192.168.1.102 255.255.255.224
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 secretpass

<output omitted>
router ospf 10
 log-adjacency-changes
 network 10.2.2.2 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

Verifying MD5 Authentication

```
R1#sho ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:31    192.168.1.102   Serial0/0/1

R1#show ip route
<output omitted>
Gateway of last resort is not set
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
O       10.2.2.2/32 [110/782] via 192.168.1.102, 00:00:37, Serial0/0/1
C       10.1.1.0/24 is directly connected, Loopback0
     192.168.1.0/27 is subnetted, 1 subnets
C       192.168.1.96 is directly connected, Serial0/0/1

R1#ping 10.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/32 ms
```

Troubleshooting Simple Password Authentication

```
Router#
debug ip ospf adj
```

- **Displays the OSPF adjacency-related events.**

```
R1#debug ip ospf adj
OSPF adjacency events debugging is on
R1#
<output omitted>
*Feb 17 18:42:01.250: OSPF: 2 Way Communication to 10.2.2.2 on Serial0/0/1,
state 2WAY
*Feb 17 18:42:01.250: OSPF: Send DBD to 10.2.2.2 on Serial0/0/1 seq 0x9B6 opt
0x52 flag 0x7 len 32
*Feb 17 18:42:01.262: OSPF: Rcv DBD from 10.2.2.2 on Serial0/0/1 seq 0x23ED
opt0x52 flag 0x7 len 32  mtu 1500 state EXSTART
*Feb 17 18:42:01.262: OSPF: NBR Negotiation Done. We are the SLAVE
*Feb 17 18:42:01.262: OSPF: Send DBD to 10.2.2.2 on Serial0/0/1 seq 0x23ED opt
0x52 flag 0x2 len 72
<output omitted>

R1#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:34    192.168.1.102   Serial0/0/1
```

Troubleshooting Simple Password Authentication Problems

Simple authentication on R1, no authentication on R2:

```
R1#
*Feb 17 18:51:31.242: OSPF: Rcv pkt from 192.168.1.102, Serial0/0/1 :
Mismatch Authentication type. Input packet specified type 0, we use type 1

R2#
*Feb 17 18:50:43.046: OSPF: Rcv pkt from 192.168.1.101, Serial0/0/1 :
Mismatch Authentication type. Input packet specified type 1, we use type 0
```

Simple authentication on R1 and R2, but different passwords:

```
R1#
*Feb 17 18:54:01.238: OSPF: Rcv pkt from 192.168.1.102, Serial0/0/1 :
Mismatch Authentication Key - Clear Text

R2#
*Feb 17 18:53:13.050: OSPF: Rcv pkt from 192.168.1.101, Serial0/0/1 :
Mismatch Authentication Key - Clear Text
```

Troubleshooting MD5 Authentication

```
R1#debug ip ospf adj
OSPF adjacency events debugging is on
<output omitted>
*Feb 17 17:14:06.530: OSPF: Send with youngest Key 1
*Feb 17 17:14:06.546: OSPF: 2 Way Communication to 10.2.2.2 on Serial0/0/1,
state 2WAY
*Feb 17 17:14:06.546: OSPF: Send DBD to 10.2.2.2 on Serial0/0/1 seq 0xB37 opt
0x52 flag 0x7 len 32
*Feb 17 17:14:06.546: OSPF: Send with youngest Key 1
*Feb 17 17:14:06.562: OSPF: Rcv DBD from 10.2.2.2 on Serial0/0/1 seq 0x32F opt
0x52 flag 0x7 len 32  mtu 1500 state EXSTART
*Feb 17 17:14:06.562: OSPF: NBR Negotiation Done. We are the SLAVE
*Feb 17 17:14:06.562: OSPF: Send DBD to 10.2.2.2 on Serial0/0/1 seq 0x32F opt
0x52 flag 0x2 len 72
*Feb 17 17:14:06.562: OSPF: Send with youngest Key 1
<output omitted>

R1#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:35    192.168.1.102   Serial0/0/1
```

Troubleshooting MD5 Authentication Problems

MD5 authentication on both R1 and R2, but R1 has key 1 and R2 has key 2, both with the same passwords:

```
R1#
*Feb 17 17:56:16.530: OSPF: Send with youngest Key 1
*Feb 17 17:56:26.502: OSPF: Rcv pkt from 192.168.1.102, Serial0/0/1 :
Mismatch Authentication Key - No message digest key 2 on interface
*Feb 17 17:56:26.530: OSPF: Send with youngest Key 1

R2#
*Feb 17 17:55:28.226: OSPF: Send with youngest Key 2
*Feb 17 17:55:28.286: OSPF: Rcv pkt from 192.168.1.101, Serial0/0/1 :
Mismatch Authentication Key - No message digest key 1 on interface
*Feb 17 17:55:38.226: OSPF: Send with youngest Key 2
```

LAB

- Configuration of Dyanamic Routing-OSPF